

Group A

Assignment No: 10

Title of the Assignment: Data Visualization III

Download the Iris flower dataset or any other dataset into a DataFrame. (e.g., <https://archive.ics.uci.edu/ml/datasets/Iris>). Scan the dataset and give the inference as:

1. List down the features and their types (e.g., numeric, nominal) available in the dataset.
 2. Create a histogram for each feature in the dataset to illustrate the feature distributions.
 3. Create a boxplot for each feature in the dataset.
 4. Compare distributions and identify outliers.
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Objective of the Assignment: Students should be able to perform the data Visualization operation using Python on any open-source dataset

Prerequisite:

1. Basic of Python Programming
2. Seaborn Library, Concept of Data Visualization.

Theory:

1. Iris Dataset:

The Iris dataset is one of the most widely used datasets in data analysis and machine learning. It was introduced by Ronald A. Fisher and contains measurements of iris flowers belonging to three species:

- Setosa
- Versicolor
- Virginica

Each observation includes four numerical features describing the flower's physical dimensions.

Importance of Data Visualization

2. Data visualization helps to:
 - Understand data distribution
 - Detect patterns and relationships

- Identify outliers and anomalies
- Support decision making in machine learning

In this experiment, two major visualization techniques are used:

- Histogram
- Boxplot

- **Histogram - Theory**

- A histogram is used to represent the frequency distribution of a numerical variable.

Key Concepts:

- X-axis → Feature values
- Y-axis → Frequency (number of observations)
- Shows shape of data:
 - Normal distribution
 - Skewness (left/right)
 - Multimodal distributions
- **Petal Length Histogram**
 - Shows 3 distinct peaks → corresponds to 3 species
 - Indicates strong separation between species
- **Sepal Width Histogram**
 - Shows overlapping values → harder to distinguish species

Inference:

- Histograms reveal that **petal features are better for classification**
- Sepal features show **less clear grouping**

Features and Their Types-

Feature Name	Type	Description
sepal_length	Numeric	Length of sepal (cm)
sepal_width	Numeric	Width of sepal (cm)
petal_length	Numeric	Length of petal (cm)
petal_width	Numeric	Width of petal (cm)

Feature Name	Type	Description
species	Nominal	Type of flower (setosa, versicolor, virginica)

Histogram for Each Feature

```
import seaborn as sns
import matplotlib.pyplot as plt

# Load dataset
iris = sns.load_dataset('iris')

# Plot histograms
iris.hist(figsize=(10,8))
plt.suptitle("Histograms of Iris Features")
plt.show()
```

Inference from Histograms

- Sepal features show normal distribution.
- Petal features show distinct clustering (helpful for classification).
- Petal length & width clearly separate species.

Boxplot for Each Feature-

```
import seaborn as sns
import matplotlib.pyplot as plt

iris = sns.load_dataset('iris')

# Boxplots for each feature
plt.figure(figsize=(10,6))
sns.boxplot(data=iris)
plt.title("Boxplot of Iris Features")
plt.show()
```

Observations-

- Sepal Length & Width
 - Moderate spread
 - Few outliers present
 - Overlapping distributions among species
- Petal Length & Width

- **Highly distinct distributions**
- **Less overlap → strong classification features**
- **Very few or no extreme outliers**
- **Outliers**
 - **Mostly seen in sepal_width**
 - **Petal features are more consistent**

Conclusion:

The Iris dataset demonstrates that petal-based features provide better separability and classification accuracy compared to sepal features, with minimal outliers and well-defined distributions.